

Professional Summary

Artificial Intelligence and Machine Learning graduate with hands-on experience in Python, machine learning, deep learning, NLP, and computer vision through academic projects and internships. Skilled in developing AI solutions, data analysis, and software development using modern tools and frameworks. Strong problem-solving abilities with experience working in collaborative development environments and a passion for learning new technologies.

Professional Experience

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| <b>AI Intern, Support Vectors AI Labs</b> <ul style="list-style-type: none"><li>Developed production-oriented AI applications using Python, LangChain, transformer models, and vector embeddings, improving semantic retrieval relevance</li><li>Built multimodal machine learning pipelines integrating SigLIP-based image-text embeddings for Computer Vision and NLP tasks across 250+ indexed image-text pairs.</li><li>Developed Retrieval-Augmented Generation (RAG) workflows and optimized inference pipelines, reducing retrieval latency while improving contextual response quality.</li><li>Designed modular AI agent workflows integrating prompt orchestration, NLP pipelines, and retrieval systems while collaborating in agile Git-based environments to develop scalable AI solutions.</li></ul> | 01/2026 – Present<br>Remote |
| <b>Web Developer Intern, Opportive</b> <ul style="list-style-type: none"><li>Engineered robust backend services utilizing the Express.js framework and MongoDB database, enhancing overall application scalability and optimizing API performance.</li><li>Developed RESTful APIs and optimized database queries, reducing server-side processing time and improving application performance.</li><li>Applied debugging, testing, and Git-based version control practices to enhance software reliability, code quality, and maintainability.</li><li>Worked collaboratively within agile development teams to proactively identify and address performance bottlenecks, delivering robust and scalable software solutions that enhanced overall system efficiency.</li></ul>                                      | 07/2025 – 08/2025<br>Remote |

Education

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| <b>Bachelor of Engineering (B.E.) in Artificial Intelligence &amp; Machine Learning,</b><br>SDM College of Engineering & Technology Dharwad<br>CGPA: 8.86 | 12/2022 – 05/2026<br>Dharwad, India |
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Skills

| Programming                                                                                                                                                  | Tools                                                                                                                          | Frameworks & Tools                                                                                               | Machine Learning & AI                                                                                                                                                     |
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| <ul style="list-style-type: none"><li>Python</li><li>SQL</li></ul>                                                                                           | <ul style="list-style-type: none"><li>Git &amp; GitHub</li><li>Jupyter Notebook</li><li>VS Code</li><li>Cursor</li></ul>       | <ul style="list-style-type: none"><li>Pandas</li><li>Numpy</li><li>FAISS</li><li>Sentence Transformers</li></ul> | <ul style="list-style-type: none"><li>Machine Learning</li><li>Supervised Learning</li><li>Unsupervised Learning</li><li>Deep Learning</li><li>NLP</li><li>LLMs</li></ul> |
| <b>AI Engineering</b> <ul style="list-style-type: none"><li>Semantic Search</li><li>Embeddings</li><li>Vector Databases</li><li>Prompt Engineering</li></ul> | <b>Soft Skills</b> <ul style="list-style-type: none"><li>Problem Solving</li><li>Collaboration</li><li>Quick learner</li></ul> |                                                                                                                  |                                                                                                                                                                           |

Technical Projects

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| <b>Person of Interest Search using Vision-Language Models</b>  <ul style="list-style-type: none"><li>Created an advanced search system that connects images and text, enabling more accurate and meaningful matches using cutting-edge visual and language technologies.</li><li>Implemented embedding-based retrieval using FAISS to support scalable similarity search across image and text representations.</li><li>Developed a scalable image retrieval system designed to efficiently handle datasets larger than 256 images.</li><li>Applied ranking and reranking strategies to improve retrieval precision and relevance.</li><li>Built a modular experimentation framework for evaluating embeddings, retrieval methods, and ranking approaches.</li></ul> | 04/2026 – 05/2026 |
| <b>Embedding Space Analysis and Fine-Tuning of Transformer Models</b>  <ul style="list-style-type: none"><li>Fine-tuned DistilBERT on the 20 Newsgroups dataset and analyzed embedding representations using PCA, t-SNE, and UMAP.</li><li>Increased Silhouette Score from 0.047 → 0.643, demonstrating significantly improved cluster separability.</li><li>Improved Linear Probe accuracy from 96% → 98% after fine-tuning.</li><li>Compared BERT and Sentence Transformer (all-MiniLM-L6-v2) embeddings to study the impact of contrastive learning on semantic structure.</li></ul>                                                                                                                                                                              | 03/2026 – 04/2026 |